CompositionMeasure for MultiDP

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `CompositionMeasure` for `MultiDP`.

1 Hoare Triple

Precondition

Compiler-verified

The output measure is derived from the common capabilities of all child measures. Only capabilities with a common composition theorem are retained.

Caller-verified

None.

Postcondition

Theorem 1.1. `composability` returns `Ok(out)` if the composition of a vector of privacy guarantees is bounded above by `compose(d_mids)` under the reported adaptivity and composability. Otherwise it returns an error.

Proof. The composition implementation evaluates each representation only when every child supplies that representation. Independently certified approximate-DP bounds compose by the basic sequential composition theorem: $\varepsilon = \sum_i \varepsilon_i$ and $\delta = \min(1, \sum_i \delta_i)$. The certificate capability is retained only when every child guarantees it; its purity is pure only when every child guarantees zero delta. No certificate is inferred from an analytical profile. Native zCDP parameters are composed by adding ρ_i and source deltas. Pure RDP parameters are composed pointwise by adding their curves; pure zCDP is embedded into RDP using the exact $\alpha\rho_i$ relation. Gaussian-DP parameters, when enabled, are composed by $\mu^2 = \sum_i \mu_i^2$.

Each accumulator uses directed interval arithmetic, so its returned value is an upper bound on the corresponding exact composition. The certificate epsilon must remain finite; an unrepresentable finite upper bound raises overflow on that path. Analytical profiles are not composed without a representation-specific theorem. If no common supported capability exists, construction fails rather than advertising an unsupported composition. Therefore every representation retained in the resulting `PrivacyGuarantee` is a valid bound for the same composed privacy relation. □

References